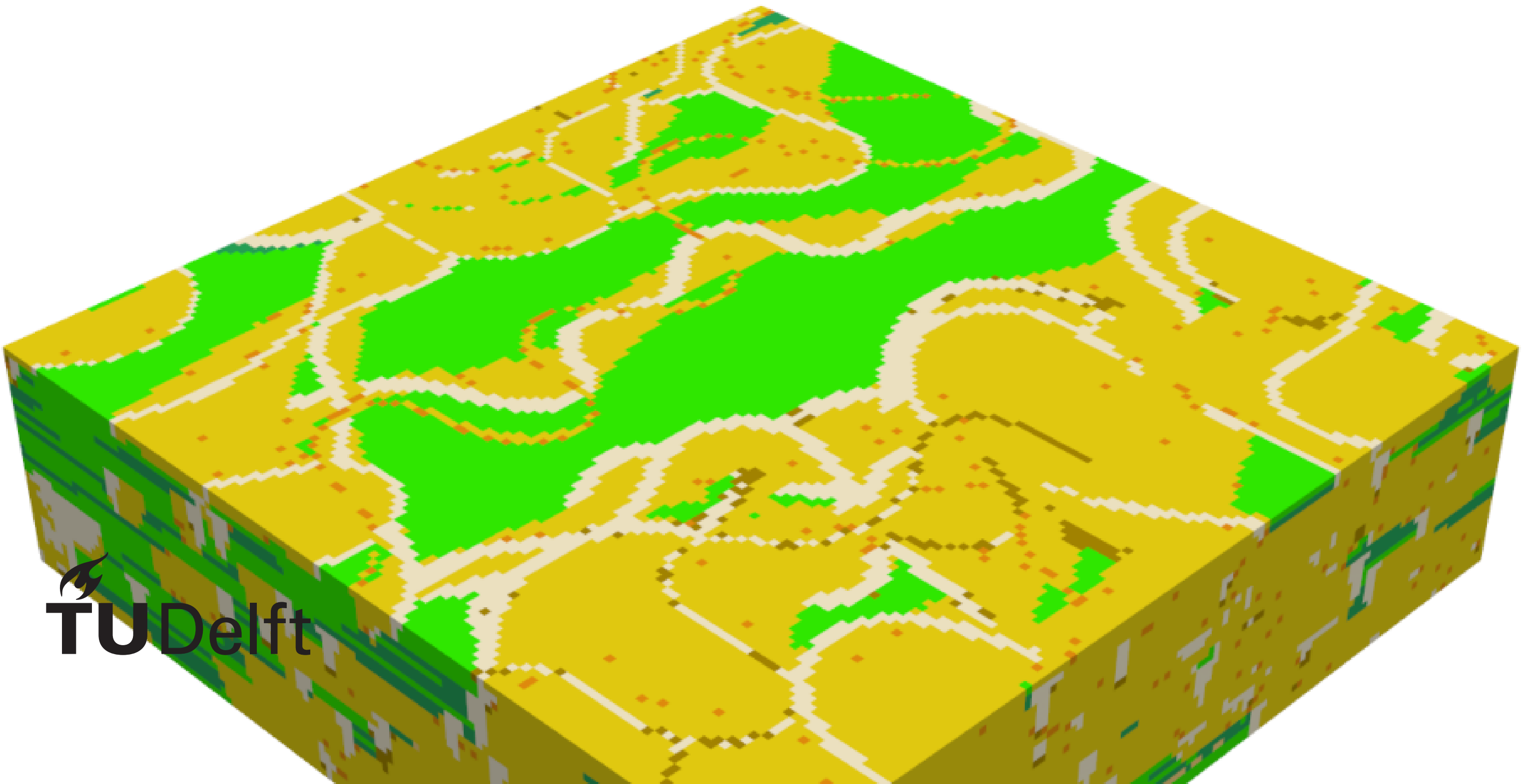
Evaluating the feasibility of Generative Adversarial networks to generate geological facies models

A Case study for the Delft Area

M.D. Torres Ruhe



Evaluating the feasibility of Generative Adversarial networks to generate geological facies models

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by

M.D. Torres Ruhe

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Abstract

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Acknowledgements

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List of Abreviations

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cGAN Conditional Generative Adversarial Network. 11

DSST Delft Sandstone. 3, 4

FluvGan Fluvial Generative Adversarial Network. 11

GAN Generative Adversarial Network. 11

TUD Technische Universiteit Delft. 3

TvD True Vertical Depth. 3

WNB West Netherlands Basin. 3

Contents

1 Introduction

Sedimentary deposits represent a vital commodity in modern society due to the various resources that can be extracted from them. These resources include minerals, groundwater, geothermal heat, and hydrocarbons (e.g., Brekke et al., 2001; Donselaar et al., 2017; Donselaar et al., 2015; Morris & Ramanaidou, 2007). Moreover, recent studies have investigated storage possibilities for carbon dioxide and hydrogen in this subsurface environment (Heinemann et al., 2021; Zhou et al., 2020). Beyond the increase in subsurface applications, the world-wide population growth and rising demand for energy and minerals required for the energy transition must be considered. Consequently, the demand for resources hosted within fluvial sedimentary deposits is expected to increase while the distribution of the commodities themselves becomes sparser and of lower quality (e.g., Calvo et al., 2016; Spittler et al., 2020). This necessitates a deeper understanding of subsurface environments to enable more precise resource management. In the exploration process, after collecting data and defining a geological conceptual model, engineers generate 3D geomodels to predict reservoir flow behavior and resource distribution (Royer et al., 2015; SONG et al., 2022).

1.1. Problem definition

The subsurface environment is characterized by a lack of data, as data acquisition projects are typically costly and time-consuming. As a result, engineers must develop an understanding of the subsurface based on limited information. This low-data environment requires the construction of 3D geomodels through interpretations of a conceptual model. The geological conceptual model represents hypotheses about the most likely rock mass configuration, constructed from sparse data (Koltermann & Gorelick, 1996). Following the definition of a conceptual model, several modelling approaches may be applied, introducing sources of uncertainty:

- **Limited input of geological prior knowledge:** When developing a geological realization, the choice of model is a critical consideration. Often, models rely on a weak geological prior, as their input is restricted to distributions of components (Rongier & Peeters, 2025b). While methods such as object modelling can recreate complex patterns, more intricate relationships are not always captured (Sun et al., 2023). Moreover, there is a need to apply a multi-conceptual model approach to minimize uncertainties associated with single-conceptual models (Enemark et al., 2019). However, developing a full-scale model for each conceptualization is time-consuming, as cases with different architectures and dimensions must be defined, making this approach labor-intensive.

- **Uncertainty quantification:** The workflow of geological modelling is divided into multiple steps, with uncertainty added at each stage, either through data inputs or the modelling steps themselves (Høyer et al., 2024). Uncertainties are generally categorized into objective and subjective types (Fookes, 1997): objective uncertainties relate to data variability, while subjective uncertainties arise from interpretations. Due to biases inherent in interpretations, these uncertainties can propagate to the final 3D model. Therefore, modelling requires professional judgment to balance objectivity and subjective intuition (Zorzi et al., 2025). As Caers (2011) noted, the uncertainty arising from subjective sources may theoretically be infinite in magnitude.

The limited geological prior knowledge and high uncertainty call for a modelling process that can capture the complexity of the subsurface while honoring available data. Consequently, the question arises as to how the current modelling workflow can be adapted so that 1) prior geological knowledge and geological complexity are better utilized and 2) uncertainty in the model is better quantified.

1.2. Research gap and opportunity

In response to these challenges in the geological modelling workflow, machine learning techniques have been proposed as a potential solution. Their main advantage is the ability to learn complex spatial relationships from data, which can be used to capture the complexity of the subsurface. Additionally, once trained and validated, a well-curated machine learning model can generate a large number of realizations efficiently, facilitating uncertainty quantification. These characteristics may help better capture geological complexity and quantify uncertainty in the modelling workflow.

The application of machine learning in geological modelling is not a new approach, as shown by Demyanov et al. (2008). However, recent developments in computer science and computing power have introduced Deep Generative Models (DGMs). Recent work in geological modelling has demonstrated the potential for several DGM techniques, such as Variational Auto-Encoders (VAEs), Generative Adversarial Networks (GANs), and 3D Latent Diffusion Models, to produce geologically realistic realizations (e.g., Bhavsar et al., 2024; Di Federico & Durlofsky, 2025; Laloy et al., 2017; Song et al., 2023). These models are advantageous because, once trained, they can generate realizations much faster than traditional process-based models and, given sufficient training data, can capture non-stationarity within the subsurface (Rongier & Peeters, 2025b).

While various DGM architectures exist, this study focuses on GANs due to their established research history in geomodelling and the wide availability of proven codebases and architectures tailored for this purpose. In contrast, while VAEs and Diffusion models show promise, their application to complex 3D reservoir conditioning is less documented in current literature. To address the low-data challenge, process-based modelling software like FLUMY (Martin et al., 2016) can be utilized to generate synthetic training datasets that represent geological prior knowledge. As Rongier and Peeters (2025b) stated: "We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs." Similarly, Di Federico and Durlofsky (2025) noted: "Finally, the application (and extension as required) of the overall methodology to real cases should be performed." With this in mind, the goal of this project is to assess the performance of deep generative modelling in real-world applications.

1.3. Objective and research questions

The main goal of this thesis is to **assess the practical feasibility of using a Generative Adversarial Network (GAN) to generate realizations of a real-world reservoir by conditioning it to well data**. The outcome of this thesis may indicate whether GANs can be used to model the subsurface in various settings and if other DGM techniques should be considered. Hence, the main research question of this work is: **To what extent can Deep Generative Modelling (DGM) be used to create geologically plausible realizations of the Delft Sandstone reservoir while conditioned to sparse real-world data?**
To help answer this question, several subquestions are formulated:

1. How should the Delft Sandstone's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modeling?
2. Which DGM methods are best suited for 3D geological modelling?
3. How well is the model able to reproduce the complexity inherent in the training data?
4. How well does the conditioned model produce geologically realistic structures and is it able to produce new, previously unseen structures?

1.4. Thesis outline

The remainder of this thesis is structured as follows:

- **Chapter 2:** Provides the theoretical background on geological modelling, deep generative models, and the geological setting of the Delft Sandstone.
- **Chapter 3:** Describes the methodology used to generate synthetic training data with FLUMY and the architecture of the GAN used for reservoir modelling.

- **Chapter 4:** Presents the results of the training process and the evaluation of the generated realizations.
- **Chapter 5:** Discusses the implications of the results, the limitations of the proposed approach, and potential future research directions.
- **Chapter 6:** Summarizes the main findings of the thesis and answers the research questions.

2 Background

This chapter establishes the theoretical and geological foundations upon which this research is built. It provides a comprehensive overview of the geological setting of the Delft reservoir and a literature review of modern geomodelling and machine learning (ML) techniques. Section 2.1 introduces the geological context of the Delft reservoir, detailing its depositional environment and current conceptual model. Section 2.2 explores the state-of-the-art in geological modelling, comparing traditional techniques and their respective limitations. Finally, Section 2.3 provides an in-depth review of machine learning and deep learning (DL) principles, focusing on their historical evolution and contemporary application to subsurface characterization. This balanced overview is intended to provide both geological and computer science readers with the necessary context to understand the research methodology and results presented in subsequent chapters.

2.1. The Delft Geothermal Reservoir

The data utilized in this study is sourced from the Delft geothermal reservoir, situated beneath the Technische Universiteit Delft (TUD) Campus in the Netherlands. As of February 2026, a geothermal doublet is actively producing heat from this reservoir (Beerda, 2026). A doublet system consists of a two-well setup: a producer well extracts warm formation water, and an injector well returns the cooled water to the reservoir. At the Delft site, the producer well (**DEL-GT-01**) was drilled to a True Vertical Depth (TvD) of approximately 2,337 m, while the injector well (**DEL-GT-02-S2**) reached a TvD of approximately 2,100 m (Vardon et al., 2024). The project serves both commercial and research objectives, acquiring extensive geophysical logs and core data, supplemented by surface and downhole monitoring instrumentation (Vardon et al., 2020; Vardon et al., 2024). Despite the high data density, the cored section is restricted to the first 80 m of the **DEL-GT-01** well (approximately one-fourth of the reservoir thickness), while **DEL-GT-02-S2** contains no cored sections. The primary target is the Delft Sandstone (DSST), a member of the Lower Cretaceous Nieuwerkerk Formation in the West Netherlands Basin (WNB) (Donselaar et al., 2015).

2.1.1. Geology of the Delft Reservoir

The WNB is a 60 km wide transtensional basin in the southwest of the Netherlands, characterized by NW-SE trending structural highs and depressions (Donselaar et al., 2015; Ziegler, 1990). The WNB entered an active rifting stage during the Middle Jurassic (Bodenhausen & Ott, 1981; Den Hartog Jager, 1996; Willems et al., 2020), leading to the formation of several half-grabens filled with terrestrial sediments sourced from neighboring regions (Den Hartog Jager, 1996). The progressive filling of these half-grabens coincided with a change in the depositional environment as a transgressive shoreline moved the system from fluvial to marine. The Nieuwerkerk Formation comprises three members: the Alblasserdam, the DSST, and the Rodenrijs Claystone.

The Alblasserdam member was deposited in a fluvial environment characterized by stacked channels and floodplain deposits (DeVault & Jeremiah, 2002). The DSST directly overlays the Alblasserdam and consists of thick, stacked fluvial-deltaic channel sandstones with high net-to-gross values, oriented NW-SE (Bruining, 2023; Groenenberg et al., 2010; Mijnlieff, 2020; Willems et al., 2020). The DSST is in turn overlain by the Rodenrijs Claystone, an organic-rich lagoonal deposit (Bruining, 2023; DeVault & Jeremiah, 2002; Willems et al., 2020).

Based on the sedimentological analysis of the **DEL-GT-01** core and **DEL-GT-02-S2** sidewall samples, three primary facies associations (FAs) have been identified (Bruining, 2023):

- **FA1: Fluvial and distributary channel deposits.** Primarily light grey, fine- to medium-grained sandstones featuring low-angle cross-bedding and ripple lamination, with unit thicknesses ranging from 5 to 80 m.

- **FA2: Crevasse splays and levee deposits.** Composed of moderately sorted, very fine- to medium-grained sandstones interbedded with dark grey siltstones, with thicknesses ranging from 1 to 5 m.

- **FA3: Overbank deposits and paleosols.** Consists of dark grey siltstones, claystones, and very fine-grained sand deposited in low-energy environments. This association is characterized by abundant organic material, slickensides, and siderite nodules.

The specific depositional environment of the DSST remains a subject of scientific discussion. It has been attributed to both fluvial-meandering (Donselaar et al., 2015; Vondrak et al., 2018; Willems et al., 2020) and fluvio-deltaic systems (Bruining, 2023; van Adrichem Boogaert & Kouwe, 1993; Weert et al., 2024). Recent analysis suggests that the cored section represents a deltaic environment (Bruining, 2023). However, due to limited core coverage, it is hypothesized that the lower DSST likely represents a fluvial environment, while the upper section reflects the transgressive shift toward a deltaic system.

2.1.2. Deltaic Depositional Environment

Deltaic environments form at the land-sea interface where river systems supply clastic sediments. Delta morphology is governed by the relative influence of fluvial, wave, and tidal processes (Kolb & Van Lopik, 1966). The subaerial delta plain is typically divided into two domains (Heldreich et al., 2017; Kolb & Van Lopik, 1966):

- **Upper Delta Plain:** Lies above the limit of saltwater intrusion. Sedimentation is driven by distributary channel migration, overbank flooding, and crevassing. Environments include meandering and braided channels, lacustrine fills, and floodplains.

- **Lower Delta Plain:** Subject to river-marine interaction and tidal influence. Channels are more numerous and exhibit bifurcating or anastomosing patterns. Interdistributary bays, bay fills (crevasse splays), and marshes dominate the landscape.

In the DSST, FA1 features organic material and siltstones indicating slackwater periods, while the lack of bioturbation suggests high sedimentation rates. The fine-grained nature of FA2 and FA3 reflects more distal positions on the delta plain. Siderite nodules in FA3 indicate anoxic, waterlogged conditions (Samuel, 1959), while soft-sediment deformation suggests rapid loading of mudstones over saturated sands. These features collectively indicate that FA3 represents deposition within a delta plain, likely in an interdistributary bay or a pro-delta environment.

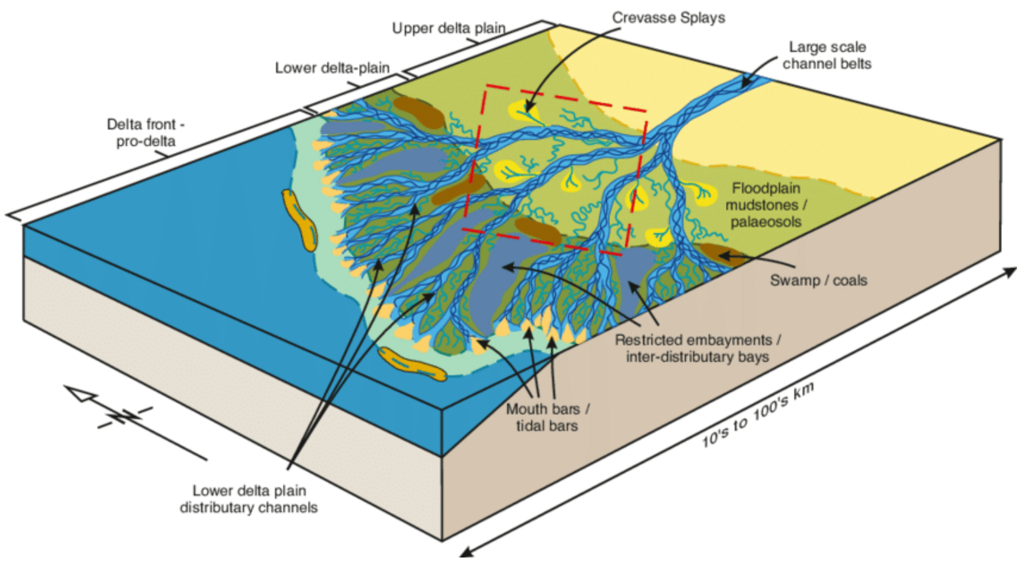

Figure 2.1: Depositional model of the Mungaroo Formation, illustrating the geometries and domains of a fluvial-deltaic system (Heldreich et al., 2017).

2.2. Geological Modelling

The primary objective of 3D reservoir modelling is to provide a representative volume of the subsurface for flow simulation and decision support. These models are dynamic, requiring updates as new data becomes available to refine predictions over the reservoir's lifetime (Emerick, 2025; Ringrose & Bentley, 2016). Existing techniques are generally categorized into three types (Koltermann & Gorelick, 1996):

- **Geostatistical (Pixel-based) Models:** Use spatial statistics (e.g., variograms) for cell-by-cell property assignment. They are efficient and honor dense well data but often fail to capture complex connectivity, resulting in pixelated realizations.
- **Object-based Models:** Directly place geometric features (e.g., channels) based on geological rules. They offer high visual realism but struggle to honor dense or closely spaced well data.
- **Process-based Models:** Simulate the physical laws of sediment transport and deposition (conservation of mass and momentum). They provide the highest degree of realism but are computationally expensive and difficult to condition to specific hard data like well logs.

To leverage the strengths of each, hybrid models have been developed (Lopez et al., 2009; Michael et al., 2010). Table 2.1 summarizes the comparison between these techniques.

Attribute	Geostatistical	Object-Based	Process-Based
Geological Prior	Variograms, training images.	Geometric shapes and rules.	Physical boundary conditions.
Realism	Low to Moderate.	Moderate to High.	Very High.
Key Advantages	Computational speed, well conditioning.	Realistic connectivity.	Physical consistency.
Key Disadvantages	Lack of crisp shapes.	Difficult conditioning.	High computational cost.

Table 2.1: Comparison of spatial modelling techniques in geomodelling.

2.3. Machine Learning

Machine learning (ML) involves the development of algorithms that learn patterns and inference from data to perform specific tasks without explicit programming (Goodfellow et al., 2016). The field has evolved significantly since Samuel (1959) introduced the term, leading to diverse techniques applied across various domains. ML is primarily categorized into:

- **Supervised Learning:** The algorithm is trained on a labeled dataset (input-output pairs).
- **Unsupervised Learning:** The algorithm identifies patterns in unlabeled data without prior knowledge of the output.

Common ML tasks include classification, regression, clustering, and generation. The modern state of ML is largely driven by Artificial Neural Networks (ANNs). Inspired by the biological structure of the brain, ANNs consist of interconnected artificial neurons that process information across multiple layers (Goodfellow et al., 2016). These networks excel at uncovering complex, non-linear relationships in vast datasets (LeCun et al., 2015).

2.3.1. Deep Learning and Inductive Biases

Deep learning (DL) is a subset of ML focusing on multi-layered neural networks that represent data with multiple levels of abstraction (Goodfellow et al., 2016; LeCun et al., 2015). Two critical concepts in DL are the "curse of dimensionality" and "inductive biases." The curse of dimensionality describes the exponential increase in data required as the dimensionality grows, making high-dimensional estimation computationally challenging.

Inductive biases are architectural choices that prioritize certain outcomes, helping models generalize from limited data (Prince, 2023). Different architectures apply specific biases:

- **Convolutional Neural Networks (CNNs):** Leverage spatial locality and translation invariance, making them ideal for image and 3D volume tasks.
- **Recurrent Neural Networks (RNNs):** Designed for sequential data to capture temporal dependencies.
- **Physics-Informed Neural Networks (PINNs):** Incorporate physical laws into the loss function to ensure consistency with scientific principles.

A critical component of the DL workflow is the use of validation and testing protocols to prevent overfitting and ensure that learned patterns are robust and applicable to unseen data.

2.3.2. Geomodelling with Deep Learning

Integration of DL into geomodelling has evolved from simple pixel-based interpolations to sophisticated structural representations. DL allows for the direct learning of geological features from large datasets or process-based simulations, bridging the gap between descriptive statistics and geological realism (Laloy et al., 2017). CNNs have been particularly successful in identifying structural features like faults or channels from seismic and well data (Sun et al., 2023). These techniques enable the generation of stochastic realizations that honor both the global geological concept and local conditioning data, enhancing uncertainty characterization (Song et al., 2023).

2.3.3. Generative Adversarial Networks

Generative Adversarial Networks (GANs) utilize a competitive framework between two networks (Goodfellow et al., 2014): a Generator (**G**) and a Discriminator (**D**). The generator creates synthetic samples, while the discriminator attempts to distinguish them from real data. This minimax game is expressed as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{D}, \mathbf{G}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})}[\log \mathbf{D}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - \mathbf{D}(\mathbf{G}(\mathbf{z})))] \tag{2.1}$$

where $\mathbf{x}$ represents real data and $\mathbf{z}$ is a latent vector from a prior distribution $p_{\mathbf{z}}$. GANs can learn to generate high-fidelity geological realizations that capture complex features without an explicit likelihood function. Conditional GANs (cGANs) allow auxiliary information (e.g., well logs) to steer the generation process (Mirza & Osindero, 2014).

Recent research has focused on improving the stability of GAN-based geomodels through refined distance metrics like the Mean Sliced Wasserstein Distance (MSWD) (Bhavsar et al., 2024). MSWD projects high-dimensional distributions onto 1D slices, providing a computationally efficient metric for training stability and geological consistency. Furthermore, techniques like latent space optimization and one-hot encoding for categorical facies data have enhanced the reliability of these models in reservoir conditioning tasks (Bhavsar et al., 2024; Rongier & Peeters, 2025b).

3 Methodology

This chapter describes the comprehensive project workflow, beginning with the generation of 3D synthetic datasets used for training, validating, and testing the models (Section 3.1). Section 3.2 details the Generative Adversarial Network (GAN) framework, including the network architectures, the facies representation strategy, and the loss functions employed during training. Finally, Section 3.2.2 introduces the metrics and simulation methods used to evaluate the geological realism and practical utility of the generated realizations.

3.1. Training Data Generation

3.1.1. Conceptual Models

Training deep learning algorithms for 3D geomodelling requires a substantial volume of ground-truth data. Given the scarcity and high uncertainty of subsurface measurements, synthetic data generation using stochastic algorithms is the established approach to provide sufficient training samples (Rongier & Peeters, 2025b; Sun et al., 2023). In this project, three distinct geological conceptual models are employed to capture the range of environments hypothesized for the Delft Sandstone (see Chapter 2):

- **Dataset A (Upper Plain Deltaic):** Characterized by meandering fluvial channels and extensive floodplain deposits.
- **Dataset B (Lower Plain Deltaic):** Reflects increased marine influence with shorter, more numerous distributary channels.
- **Dataset C (General Deltaic):** A stochastic combination of both settings to represent a transitional depositional system.

3.1.2. Dataset Descriptions

The training datasets were generated using FLUMY (Martin et al., 2016), a process-based stochastic modelling software designed to reproduce meandering channelized reservoirs. FLUMY simulates facies, relative grain size, and depositional time for each voxel in a 3D grid based on specified hydraulic and sedimentological rules. The models are defined on a reservoir scale with a grid resolution of $128 \times 128 \times 32$ voxels. With a voxel dimension of 20 m $\times$ 20 m $\times$ 1 m, the total simulated volume covers 2560 m $\times$ 2560 m $\times$ 32 m.

The geological variability between the three datasets is controlled by three primary NEXUS input parameters: Net-to-Gross (NTG), maximum channel depth, and maximum sand body extension. The specific configurations are summarized in Table 3.1. While Chapter 2 notes that individual FA1 units can reach 80 m in thickness, the 32 m grid depth used here represents a focused reservoir interval where the conditioning challenge is most acute.

Parameter	**Dataset A**	**Dataset B**	**Dataset C**
Net-to-Gross (%)	67	67	67
Max. Channel Depth (m)	6	4	4–6
Max. Sand Body Extension	100	60	60–100

Table 3.1: NEXUS input parameters for the three training datasets.

A total of 20,000 samples were generated for each dataset. Each sample is represented using 9 primary facies classes relevant to the fluvial-deltaic system, as shown in Table 3.2.

Value	Facies Key	Color	Description
0	Background		Undefined/Muddy matrix
1	Channel Lag		Coarse active channel fill
2	Point Bar		Meandering lateral accretion
3	Sand Plug		Fine-grained channel abandonment
4	Crevasse Splay		Proximal high-energy overbank
5	Crevasse Channel		Feeder for splay deposits
6	Levee		Bordering channel ridges
7	Overbank		Vegetated floodplain silt
8	Wetland/Paleosol		Organic-rich anoxic sediments

Table 3.2: Overview of the 9 facies classes used in the training datasets.

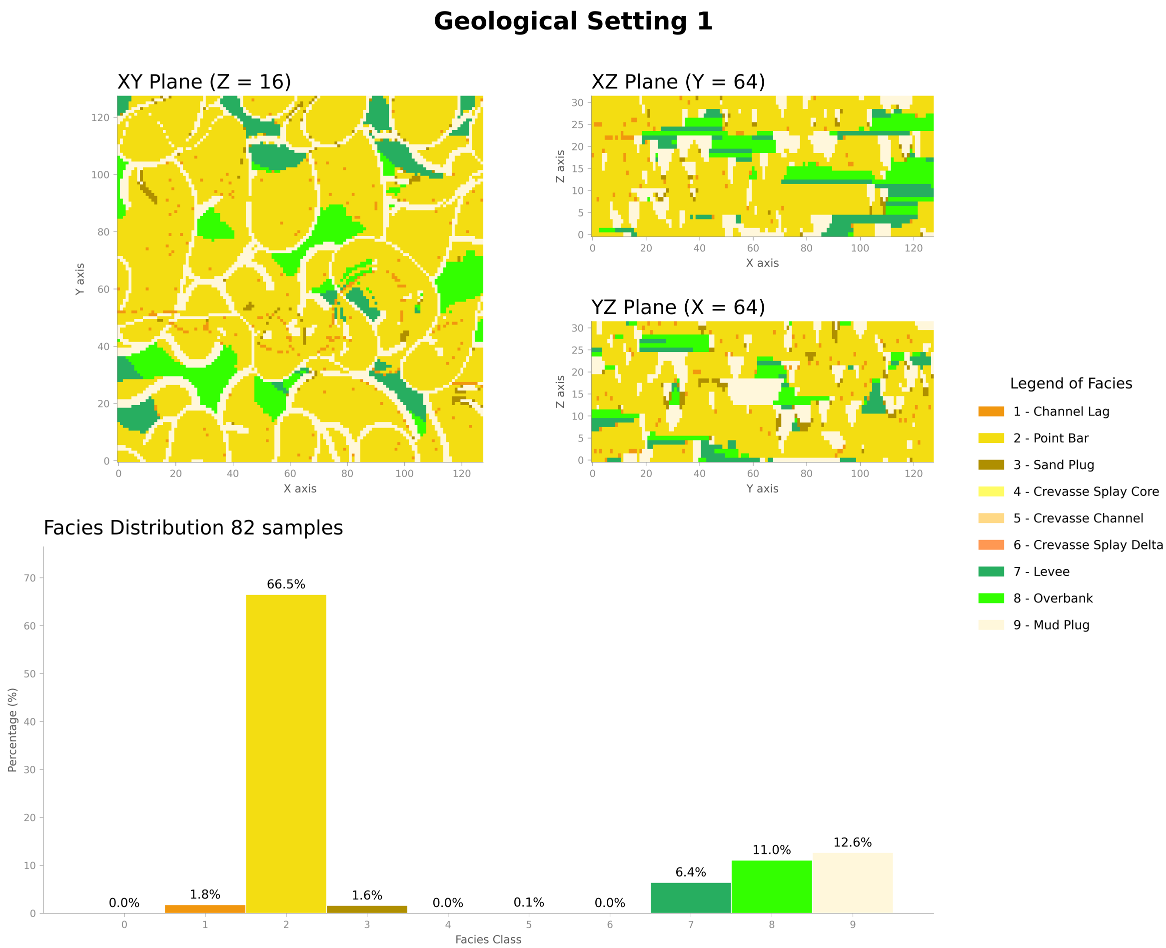

Figure 3.1: Visualization of training data from Dataset A. The rows represent facies distribution, relative grain size, and depositional time, respectively.

3.2. Fluvgan Framework

The Fluvial Generative Adversarial Network (FluvGan) model, introduced by Rongier and Peeters (2025a), is a conditional GAN architecture optimized for 3D fluvial facies modelling. It builds upon the BigGAN framework (Brock et al., 2019) to handle the high resolution and complex connectivity required for reservoir models. Previous studies have demonstrated that FluvGan can generate geologically plausible realizations while remaining training-stable (Rongier & Peeters, 2025b). Crucially, Rongier and Peeters (2025c) showed that the model's latent space can be inverted and optimized to honor hard constraints from well data.

3.2.1. Model Architecture

The architecture comprises a Generator (**G**) and a Discriminator (**D**) using 3D Residual blocks. To handle the categorical nature of facies data, the project employs a one-hot encoding strategy where each facies is represented as a separate channel in the output volume.

The generator transforms a 100-dimensional latent vector **z** into a $9 \times 32 \times 128 \times 128$ volume. It utilizes a 3D transposed convolution to project the latent vector into a $1024 \times 4 \times 4 \times 4$ feature map, followed by five residual upscaling blocks. To accommodate the anisotropy of the grid, spatial dimensions are doubled in each block (up to 128), while the depth dimension is doubled only in the first three blocks (up to 32). The final layer uses a Softmax activation to ensure the 9 facies channels sum to one per voxel.

The discriminator acts as a binary classifier, mapping the $9 \times 32 \times 128 \times 128$ input back to a single scalar logit. It consists of five residual downscaling blocks that iteratively reduce the spatial resolution while increasing the channel count to 1024. Both networks employ spectral normalization and LeakyReLU activations to ensure stable training. The generator and discriminator architectures contain approximately 11.6 and 2.0 million trainable parameters, respectively.

Hyperparameter	**Value**	**Description**
Latent Dimension ($dim(\mathbf{z})$)	100	Gaussian prior
Batch Size	16	Samples per iteration
Learning Rate (G, D)	$10^{-4}, 4 \cdot 10^{-4}$	Two-timescale update rule
Kernel Size	$3 \times 3 \times 3$	3D Convolutional filters
Optimizer	Adam	$\beta_1 = 0, \beta_2 = 0.999$

Table 3.3: Key hyperparameters for the Fluvgan architecture.

3.2.2. Conditioning and Evaluation

To honor sparse well data, a latent space optimization approach is used. The latent vector **z** is iteratively refined to minimize the difference between the generated facies at well locations and the true observations. The realism of the final realizations is evaluated using the Mean Sliced Wasserstein Distance (MSWD) across 1,000 projections, ensuring that the multi-point statistics and connectivity of the training data are preserved (Bhavsar et al., 2024). Additionally, connectivity analysis and (heat)flow simulations are performed to assess the practical impact of the modelling approach on reservoir performance predictions.

4 Results

4.1. Post Training Results

4.2. Post Conditioning Results

5 Discussion & Recommendations

5.1. Interpretations of key results

5.2. Limitations of the current work

5.3. Recommendations for future work

6 Conclusion

6.1. Answers to Research Questions

6.2. Final Remarks

Bibliography

Beerda, E. (2026). Innovatie: Delftse aardwarmtecentrale levert ook razend interessante onderzoeksgegevens [Gewijzigd op 11 maart 2026]. *Cobouw*. Retrieved March 20, 2026, from https://www.cobouw.nl/335189/innovatie-delftse-aardwarmtecentrale-levert-ook-razend-interessante-onderzoeksgegevens

Bhavsar, F., Desassis, N., Ors, F., & Romary, T. (2024). A stable deep adversarial learning approach for geological facies generation. *Computers & Geosciences*, *190*, 105638.

Bodenhausen, J., & Ott, W. (1981). Habitat of the Rijswijk oil province, onshore, the Netherlands. In *Petroleum geology of the continental shelf of North-West Europe* (pp. 301–309). Institute of Petroleum London.

Brekke, H., Sjulstad, H. I., Magnus, C., & Williams, R. W. (2001). Sedimentary environments offshore Norway—an overview. *Norwegian petroleum society special publications*, *10*, 7–37.

Brock, A., Donahue, J., & Simonyan, K. (2019). Large scale GAN training for high fidelity natural image synthesis. *Proceedings of the International Conference on Learning Representations (ICLR)*.

Bruining, J. K. (2023). *Sedimentological analysis of the well-logs, cores, plugs, and sidewall cores of the DEL-GT wells* [Master's thesis]. Delft University of Technology. https://repository.tudelft.nl/

Caers, J. (2011). *Modeling uncertainty in the earth sciences*. John Wiley & Sons.

Calvo, G., Mudd, G., Valero, A., & Valero, A. (2016). Decreasing ore grades in global metallic mining: A theoretical issue or a global reality? *Resources*, *5*(4), 36.

Demyanov, V., Pozdnoukhov, A., Kanevski, M., & Christie, M. (2008). Geomodelling of a fluvial system with semi-supervised support vector regression. *Proceedings of International geostatistics congress, Santiago, Chile*, 627–636.

Den Hartog Jager, D. (1996). Fluviomarine sequences in the lower Cretaceous of the West Netherlands Basin: Correlation and seismic expression. *Geology of Gas and Oil under the Netherlands: Selection of papers presented at the 1983 Iternational Conference of the American Association of Petroleum Geologists, held in The Hague*, 229–241.

DeVault, B., & Jeremiah, J. (2002). Tectonostratigraphy of the Nieuwerkerk Formation (Delfland Subgroup), West Netherlands Basin. *AAPG bulletin*, *86*(10), 1679–1707.

Di Federico, G., & Durlofsky, L. J. (2025). Three-dimensional latent diffusion models for parameterizing and history matching facies systems under hierarchical uncertainty. *Mathematical Geosciences*, 1–33.

Donselaar, M. E., Bhatt, A. G., & Ghosh, A. K. (2017). On the relation between fluvio-deltaic flood basin geomorphology and the wide-spread occurrence of arsenic pollution in shallow aquifers. *Science of the total environment*, *574*, 901–913.

Donselaar, M. E., Groenenberg, R. M., & Gilding, D. T. (2015). Reservoir geology and geothermal potential of the Delft Sandstone Member in the West Netherlands Basin. *Proceedings World Geothermal Congress 2015*, 1–9.

Emerick, A. A. (2025). *Ensemble data assimilation applied to geological reservoir models*. Petrobras.

Enemark, T., Peeters, L. J., Mallants, D., & Batelaan, O. (2019). Hydrogeological conceptual model building and testing: A review. *Journal of Hydrology*, *569*, 310–329. https://doi.org/https://doi.org/10.1016/j.jhydrol.2018.12.007

Fookes, P. (1997). Geology for engineers: The geological model, prediction and performance. *Quarterly Journal of Engineering Geology and Hydrogeology*, *30*(4), 293–424.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning* [http://www.deeplearningbook.org]. MIT Press.

Goodfellow, I., Pouget-Abadie, J. b., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. *Advances in neural information processing systems*, 2672–2680.

Groenenberg, R., Gilding, D., Donselaar, M., Drost, G., Korenromp, M., Loerakker, M., Wiggers, C., Wever, A., & Wolf, K. (2010). Targeting for geothermal energy production–reservoir characterization and geothermal potential of the Delft Sandstone. *72nd EAGE Conference and Exhibition incorporating SPE EUROPEC 2010*, cp–161.

Heinemann, N., Alcalde, J., Miocic, J. M., Hangx, S. J., Kallmeyer, J., Ostertag-Henning, C., Hassanpouryouzband, A., Thaysen, E. M., Strobel, G. J., Schmidt-Hattenberger, C., et al. (2021). Enabling large-scale hydrogen storage in porous media–the scientific challenges. *Energy & environmental science*, *14*(2), 853–864.

Heldreich, G., Redfern, J., Legler, B., Gerdes, K., & Williams, B. (2017). Challenges in characterizing subsurface paralic reservoir geometries: A detailed case study of the Mungaroo Formation, North West Shelf, Australia. *Geological Society, London, Special Publications*, *444*, 221–256.

Høyer, A., Sandersen, P., Andersen, L., Madsen, R., Mortensen, M., & Møller, I. (2024). Evaluating the chain of uncertainties in the 3D geological modelling workflow. *Engineering Geology*, *343*, 107792.

Kolb, C. R., & Van Lopik, J. R. (1966). Depositional environments of the Mississippi River deltaic plain—southeastern Louisiana.

Koltermann, C. E., & Gorelick, S. M. (1996). Heterogeneity in sedimentary deposits: A review of structure-imitating, process-imitating, and descriptive approaches. *Water Resources Research*, *32*(9), 2617–2658.

Laloy, E., Hérault, R., Lee, J., Jacques, D., & Linde, N. (2017). Inversion using a new low-dimensional representation of complex binary geological media based on a deep neural network. *Advances in water resources*, *110*, 387–405.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *nature*, *521*(7553), 436–444.

Lopez, S., Cojan, I., Rivoirard, J., & Galli, A. (2009). Process-based stochastic modelling: Meandering channelized reservoirs. *Analogue and numerical modelling of sedimentary systems: from understanding to prediction, Wiley, Oxford, UK*, 139–144.

Martin, L., Cojan, I., Ors, F., & Rivoirard, J. (2016, April). *Potential of FLUMY, a Meandering Fluvial Process-based Model, to Simulate Submarine Channels*. https://doi.org/10.3997/2214-4609.201600383

Michael, H., Li, H., Boucher, A., Sun, T., Caers, J., & Gorelick, S. (2010). Combining geologic-process models and geostatistics for conditional simulation of 3-D subsurface heterogeneity. *Water Resources Research*, *46*(5).

Mijnlieff, H. F. (2020). Introduction to the geothermal play and reservoir geology of the Netherlands. *Netherlands Journal of Geosciences*, *99*, e2.

Mirza, M., & Osindero, S. (2014). Conditional generative adversarial nets. *arXiv preprint arXiv:1411.1784*.

Morris, R., & Ramanaidou, E. (2007). Genesis of the channel iron deposits (CID) of the Pilbara region, Western Australia. *Australian Journal of Earth Sciences*, *54*(5), 733–756.

Prince, S. J. (2023). *Understanding deep learning*. MIT press.

Ringrose, P., & Bentley, M. (2016). *Reservoir model design* (Vol. 2). Springer.

Rongier, G., & Peeters, L. (2025a). FluvGAN: Python scripts to test generating and inverting fluvial deposits using GANs. https://doi.org/10.4121/3469c879-f443-4fcf-83a2-b6df56e96714.v1

Rongier, G., & Peeters, L. (2025b). Towards geological inference with process-based and deep generative modeling, part 1: Training on fluvial deposits. *arXiv preprint arXiv:2510.14445*.

Rongier, G., & Peeters, L. (2025c). Towards geological inference with process-based and deep generative modeling, part 2: Inversion of fluvial deposits and latent-space disentanglement. *arXiv preprint arXiv:2510.17478*.

Royer, J., Mejia, P., Caumon, G., & Collon, P. (2015). 3D and 4D geomodelling applied to mineral resources exploration—an introduction. In *3D, 4D and predictive modelling of major mineral belts in Europe* (pp. 73–89). Springer.

Samuel, A. L. (1959). Some studies in machine learning using the game of checkers. *IBM Journal of research and development*, *3*(3), 210–229.

SONG, S., SHI, Y., & HOU, J. (2022). Review of a generative adversarial networks (GANs)-based geomodelling method. *Petroleum Science Bulletin*, *7*(1), 34–49.

Song, S., Zhang, D., Mukerji, T., & Wang, N. (2023). GANSim-surrogate: An integrated framework for stochastic conditional geomodelling. *Journal of Hydrology*, *620*, 129493.

Spittler, N., Shafiei, E., Davidsdottir, B., & Juliusson, E. (2020). Modelling geothermal resource utilization by incorporating resource dynamics, capacity expansion, and development costs. *Energy*, *190*, 116407.

Sun, C., Demyanov, V., & Arnold, D. (2023). Geological realism in fluvial facies modelling with GAN under variable depositional conditions. *Computational Geosciences*, *27*(2), 203–221.

van Adrichem Boogaert, H. A., & Kouwe, W. (1993). Stratigraphic nomenclature of the Netherlands, revision and update by RGD and NOGEPA.

Vardon, P., Bruhn, D., Steiginga, A., Cox, B., Abels, H., Barnhoorn, A., Drijkoningen, G., Slob, E., & Wapenaar, K. (2020). A geothermal well doublet for research and heat supply of the TU Delft Campus. *arXiv preprint arXiv:2003.11826*.

Vardon, P., Laumann, S., van Esser, B., Ursem, L., van Schravendijk, B., Meleza, L. V., Barnhoorn, A., Abels, H., Vondrak, A., Drijkoningen, G., et al. (2024). End-of-well science programme report DEL-GT-01 and DEL-GT-02: TU Delft campus geothermal project.

Vondrak, A., Donselaar, M. E., & Munsterman, D. (2018). Reservoir architecture model of the Nieuwerkerk Formation (early Cretaceous, West Netherlands Basin): Diachronous development of sand-prone fluvial deposits. *Geological Society, London, Special Publications*, *471*, 147–171.

Weert, A., Ogata, K., Vinci, F., Leo, C., Bertotti, G., Amory, J., & Tavani, S. (2024). Multiple phase rifting and subsequent inversion in the West Netherlands Basin: Implications for geothermal reservoir characterization. *Solid Earth*, *15*(2), 121–141.

Willems, C. J., Vondrak, A., Mijnlieff, H. F., Donselaar, M. E., & Van Kempen, B. M. (2020). Geology of the upper Jurassic to lower Cretaceous geothermal aquifers in the West Netherlands Basin–an overview. *Netherlands Journal of Geosciences*, *99*, e1.

Zhou, Q., Yang, X., Zhang, R., Hosseini, S. A., Ajo-Franklin, J. B., Freifeld, B. M., Daley, T. M., & Hovorka, S. D. (2020). Dynamic processes of CO2 storage in the field: 1. multiscale and multipath channeling of CO2 flow in the hierarchical fluvial reservoir at Cranfield, Mississippi. *Water Resources Research*, *56*(2), e2019EF001360.

Ziegler, P. A. (1990). *Geological atlas of western and central Europe* (2nd, Vol. 2). Geological Society Publishing House.

Zorzi, L., Elmo, D., Tafwidla, F., Lenauer, I., & Furney, S. (2025). 3D geological models: A reliable depiction of nature or a sketch of our biases? *SSIM 2025: Fourth International Slope Stability in Mining Conference*, 15–28.